



Retail Sales Forecasting & Markdown Impact Analysis

➤ Objective:

Analyze historical retail sales data to understand:

- Which stores and departments perform best and worst?
- How markdowns and holidays influence sales?
- How economic factors like CPI and fuel prices correlate with performance?
- Which store types and sizes yield higher returns?
- What strategies to adopt for maximizing sales impact?

➤ Datasets Overview:

Three datasets (CSV format):

1. sales_data-set.csv

- **Store** – Store number
- **Dept** – Department number
- **Date** – Weekly date
- **Weekly_Sales** – Sales value
- **IsHoliday** – Boolean (is this a holiday week?)

2. features_data-set.csv

- **Store** – Store number
- **Date** – Weekly date
- **Temperature**
- **Fuel_Price**
- **MarkDown1-5** – Promotional data
- **CPI** – Consumer Price Index
- **Unemployment**
- **IsHoliday**



3. stores_data-set.csv

- **Store** – Store number
- **Type** – Store type (A, B, C)
- **Size** – Store size (sqft)

Merge Key: All datasets are connected via Store and Date.

Proposed Combined Table Name: retail_combined_sales

Project Scope

➤ Data Engineering (SQL):

- Merge the three datasets into one unified table.
- Clean null values in markdowns using logical imputation or mean substitution.
- Convert holiday data to flags and descriptions (e.g., Thanksgiving, Christmas).
- Engineer new features:
 1. **Adjusted_Sales** = Weekly_Sales adjusted with markdown effect.
 2. **Sales_per_sqft** = Weekly_Sales / Store_Size
 3. **Sales_Category** = CASE WHEN Weekly_Sales > X THEN 'High' ELSE 'Low' END
- Convert dates into WEEK, MONTH, and YEAR columns.

Analysis Questions (SQL Tasks)

➤ Sales Performance Insights

- Top 5 and bottom 5 stores by total sales.
- Top departments per store type.
- Yearly sales trend (2010, 2011, 2012) across all stores.

➤ **Markdown Impact**

- Compare average sales during markdown weeks vs non-markdown weeks.
- Identify departments that respond best to markdown campaigns.
- Create a rolling 3-week sales average to measure campaign effectiveness.

➤ **Holiday Impact Analysis**

- Rank holidays by total sales impact.
- Sales uplift (%) during holiday weeks vs average weeks.
- Department performance comparison on holidays vs regular weeks.

➤ **Store Type & Region Analysis**

- Sales per store type (A, B, C).
- Correlation of store size and sales.
- High performing stores per 1000 sqft.

➤ **Forecast & Consistency Metrics**

- Use LAG() to compute WoW sales changes.
- Identify most consistent stores using STDDEV() on weekly sales.
- Detect outlier weeks using Z-score logic.



Advanced Use Cases

- Use **Window Functions** to calculate running sales totals, ranks, and rolling averages.
- Use **CTEs** to clean and transform markdown data.
- Use **Views** for reusable datasets like HolidaySalesSummary and MarkdownAdjustedSales.
- Use **Subqueries** to compare year-over-year changes at a department level.



➤ Final Deliverables

Components	Description
SQL Script	For table creation, cleaning, analysis
Visual Report (Optional)	Reusable datasets for summary dashboards
Documentation	Steps taken, assumptions, summary tables
Summary Report	Business insights, markdown recommendations, holiday strategy